

Proof. The “either” condition can be stated as

$$\exists \begin{pmatrix} \mathbf{t} \\ \mathbf{u} \end{pmatrix} \geq \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix} : \quad \begin{pmatrix} \mathbb{1} & \mathbf{0} \\ V & Y \end{pmatrix} \begin{pmatrix} \mathbf{t} \\ \mathbf{u} \end{pmatrix} = \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix}$$

which by version II of the Farkas lemma is equivalent to

$$\begin{aligned} \stackrel{\text{FL II}}{\Longleftrightarrow} \quad & \nexists (\alpha, -\mathbf{a}) \in (\mathbb{R}^{d+1})^* : \quad (\alpha, -\mathbf{a}) \begin{pmatrix} \mathbb{1} & \mathbf{0} \\ V & Y \end{pmatrix} \geq (\mathbf{0}, \mathbf{0}), \\ & (\alpha, -\mathbf{a}) \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} < 0 \\ \Longleftrightarrow \quad & \nexists (\alpha, -\mathbf{a}) \in (\mathbb{R}^{d+1})^* : \quad \alpha \mathbb{1} - \mathbf{a}V \geq \mathbf{0}, \quad \mathbf{a}Y \leq \mathbf{0}, \quad \mathbf{a}\mathbf{x} > \alpha, \end{aligned}$$

which is equivalent to the negation of the “or” condition. $\square$

1.5 Recession Cone and Homogenization

Using the Farkas lemma, we can give an invariant description of some very important constructions (notably the recession cone and the homogenization of a convex set) and establish their basic properties. In Proposition 1.14 we will see that the homogenization $\text{homog}(P)$ of a polyhedron coincides with the “associated cone” $C(P)$ that we used in Section 1.1.

Definition 1.11. Let $P \subseteq \mathbb{R}^d$ be a convex set. Then the *lineality space* of P is defined as

$$\text{lineal}(P) := \{\mathbf{y} \in \mathbb{R}^d : \mathbf{x} + t\mathbf{y} \in P \text{ for all } \mathbf{x} \in P, t \in \mathbb{R}\},$$

and the *recession cone* (or *characteristic cone*) of P is defined as

$$\text{rec}(P) := \{\mathbf{y} \in \mathbb{R}^d : \mathbf{x} + t\mathbf{y} \in P \text{ for all } \mathbf{x} \in P, t \geq 0\}.$$

Directly from the definition we can derive that $\text{lineal}(P)$ is a linear subspace of $\mathbb{R}^d$. If we choose a complementary subspace U to $\text{lineal}(P)$ (i.e., $U \cap \text{lineal}(P) = \{\mathbf{0}\}$ and $U + \text{lineal}(P) = \mathbb{R}^d$), then P can be decomposed as the Minkowski sum

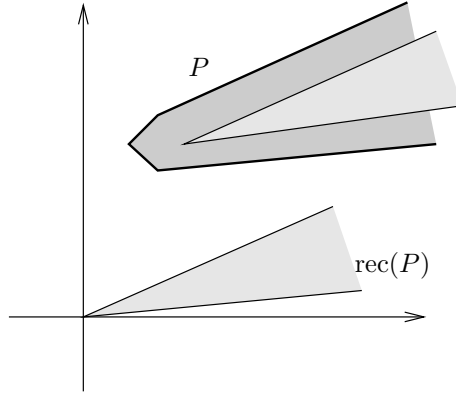
$$P = \text{lineal}(P) + (P \cap U)$$

of a linear subspace L and a convex set $P \cap U$ whose lineality space is zero: $\text{lineal}(P \cap U) = \{\mathbf{0}\}$.

This reduction usually makes it possible to consider only polyhedra with lineality space $\{\mathbf{0}\}$, known as *pointed polyhedra* (if they are nonempty).

For $\mathcal{H}$ -polyhedra we compute $\text{lineal}(P(A, \mathbf{z})) = \{\mathbf{x} \in \mathbb{R}^d : A\mathbf{x} = \mathbf{0}\}$. So, except for a trivial linear summand we can usually consider polyhedra $P(A, \mathbf{z}) \subseteq \mathbb{R}^d$ for which A has full rank d .

Similarly, we see that $\text{rec}(P)$ is a convex cone: it contains $\mathbf{0}$, any positive multiple of a vector, and any convex combination of any two of its vectors.



Proposition 1.12. *Let $P \subseteq \mathbb{R}^d$ be a convex set.*

- (i) *If $P = P(A, \mathbf{z})$ is an $\mathcal{H}$ -polyhedron, then so is its recession cone:*

$$\text{rec}(P) = P(A, \mathbf{0}).$$

- (ii) *If $P = \text{conv}(V) + \text{cone}(Y)$ is a $\mathcal{V}$ -polyhedron, then so is its recession cone:*

$$\text{rec}(P) = \text{cone}(Y).$$

Proof. Both parts are “clear,” aren’t they? Not quite: on close inspection we see that in part (ii) the direction $\text{rec}(P) \subseteq \text{cone}(Y)$ is not entirely obvious; it needs the Farkas lemma. Using version IV (for $V = \mathbf{0}$), we see that if $\mathbf{y} \notin \text{cone}(Y)$, then there exists a linear functional $\mathbf{a}$ with $\mathbf{a}Y \leq \mathbf{0}$ and $\mathbf{a}\mathbf{y} > 0$.

Now consider some $\mathbf{z} = V\mathbf{t} + Y\mathbf{u} \in \text{conv}(V) + \text{cone}(Y)$, with $\mathbf{t}, \mathbf{u} \geq \mathbf{0}$, $\mathbb{1}\mathbf{t} = 1$. For this we get

$$\mathbf{a}\mathbf{z} = \mathbf{a}V\mathbf{t} + \mathbf{a}Y\mathbf{u} \leq \mathbf{a}V\mathbf{t} = \sum_{1 \leq i \leq n} t_i \mathbf{a}\mathbf{v}_i \leq \max_{1 \leq i \leq n} \mathbf{a}\mathbf{v}_i =: K,$$

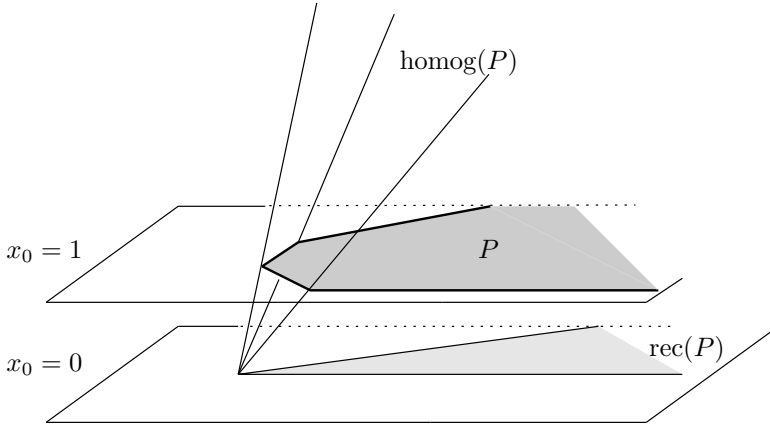
where K only depends on $\mathbf{a}$ and V . However, we get that $\mathbf{a}(\mathbf{z} + t\mathbf{y}) = \mathbf{a}\mathbf{z} + t(\mathbf{a}\mathbf{y}) \rightarrow +\infty$ for $t \rightarrow +\infty$, so we have $\mathbf{z} + t\mathbf{y} \notin P$ for t large enough, and thus $\mathbf{y} \notin \text{rec}(P)$. $\square$

Definition 1.13. Let $P \subseteq \mathbb{R}^d$ be a convex set. Then the *homogenization* of P is defined as

$$\text{homog}(P) := \left\{ t \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} : \mathbf{x} \in P, t > 0 \right\} + \left\{ \begin{pmatrix} 0 \\ \mathbf{y} \end{pmatrix} : \mathbf{y} \in \text{rec}(P) \right\}.$$

Again it is quite easy to see that for every convex set P , the homogenization $\text{homog}(P)$ is a convex cone in $\mathbb{R}^{d+1}$. Furthermore, any P can be easily recovered from its homogenization (if we don't mess with the coordinate system) as

$$P = \{\mathbf{x} \in \mathbb{R}^d : \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} \in \text{homog}(P)\}.$$



Proposition 1.14. *Let $P \subseteq \mathbb{R}^d$ be a convex set.*

- (i) *If $P = P(A, \mathbf{z})$ is an $\mathcal{H}$ -polyhedron, then its homogenization is also an $\mathcal{H}$ -polyhedron:*

$$\text{homog}(P) = P\left(\begin{pmatrix} -1 & \mathbf{0} \\ -\mathbf{z} & A \end{pmatrix}, \begin{pmatrix} 0 \\ \mathbf{0} \end{pmatrix}\right) = C(P).$$

- (ii) *If $P = \text{conv}(V) + \text{cone}(Y)$ is a $\mathcal{V}$ -polyhedron, then so is its homogenization:*

$$\text{homog}(P) = \text{cone}\left(\begin{pmatrix} \mathbf{1} & \mathbf{0} \\ V & Y \end{pmatrix}\right) = C(P).$$

Proof. This now follows from Proposition 1.12. □

1.6 Carathéodory's Theorem

The following proposition states two versions (linear and affine) of another basic tool, known as Carathéodory's theorem. We want to emphasize that in contrast to the Farkas lemma this is completely elementary and also computationally quite trivial. However, it can be successfully applied, for example, to sharpen the Farkas lemmas, as well as the main theorems and representation theorems for polytopes, cones, and polyhedra; see the next lecture.